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Unit – 1 : Session – 5 : SLO – 2 | Signed magnitude subtraction

■ Multiple Choice Questions (5)

1. b — Sign
2. b — -2 ((+3)-(+5): smaller minus larger → negative, magnitude $5-3=2$)
3. c — (+A) - (-B) (becomes addition of magnitudes, which can overflow)
4. b — +2 ((-3)-(-5) = +2: magnitudes $5-3=2$, sign of larger magnitude operand is +)
5. b — -7 to +7

■ Two-Mark Questions (2)

1. (+A)-(-B) and (-A)-(+B) turn into magnitude ADDITION (subtracting a negative). The result magnitude A+B can exceed the (N-1)-bit magnitude field (e.g. $5+3=8 > 7$ in 4-bit), so the true result cannot be represented — overflow.
2. Same-sign subtraction steps: (i) compare magnitudes; (ii) subtract the smaller magnitude from the larger; (iii) keep the common sign if the first operand's magnitude is larger, otherwise take the opposite sign; (iv) if magnitudes are equal the result is +0.

■ Scenario-Based Long Answer (1)

- (-5) - (+3): different signs → this becomes magnitude addition: $5 + 3 = 8$, and the sign of the result is the sign of the first operand (negative) → true result -8.
- Step-by-step: -5 = 1101, +3 = 0011. Magnitudes $101 + 011 = 1000$ (8). Magnitude field is 3 bits (max 111 = 7). $8 > 7$ → magnitude cannot fit.
- OVERFLOW occurs: -8 is outside the 4-bit signed magnitude range (-7 to +7).

■ Match the Following (5)

- A. MSB — 1. Sign bit
B. (+A) - (-B) — 2. Addition; possible overflow
C. (-A) - (+B) — 3. Addition; same sign result
D. (-3) - (-5) — 4. +2 (result)
• E. 4-bit signed magnitude — 5. Range from -7 to +7

■ Fill in the Blanks (5)

1. MSB (most significant / sign) 2. subtract/compare 3. (N-1)-bit magnitude 4. +8 (overflow) 5. less efficient (harder/more complex)

■ Scenario-Based Small Activity (1)

- (+5) - (+3): both positive. Compare magnitudes: $101 > 011$. Subtract: $101 - 011 = 010$ (2). Sign stays positive.
- Final result: $0010 = +2$.

■ True or False (5)

1. FALSE — MSB stores the sign, not magnitude. 2. TRUE
3. TRUE — $(-5)-(-3) = -5+3 = -2$. 4. FALSE — $(+3)-(+5) = -2$, not $+2$.
5. FALSE — Two's complement has the simpler hardware.